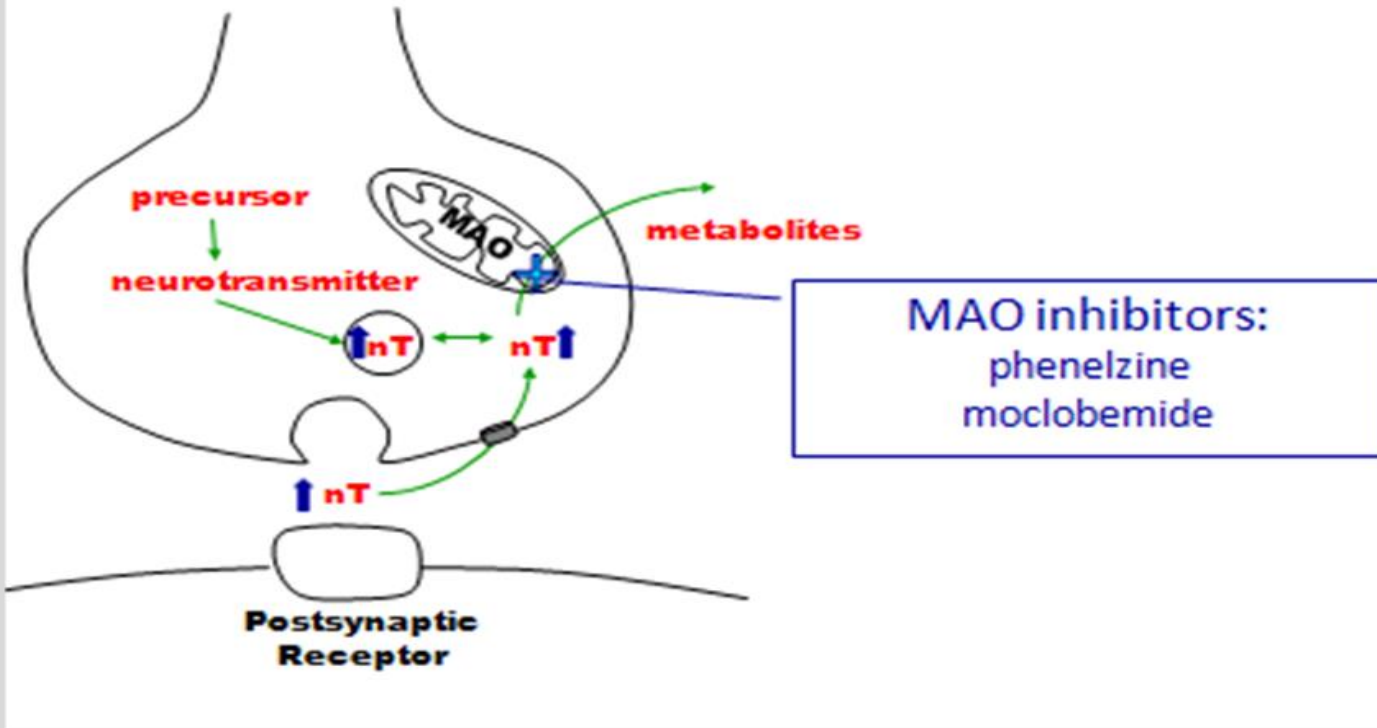


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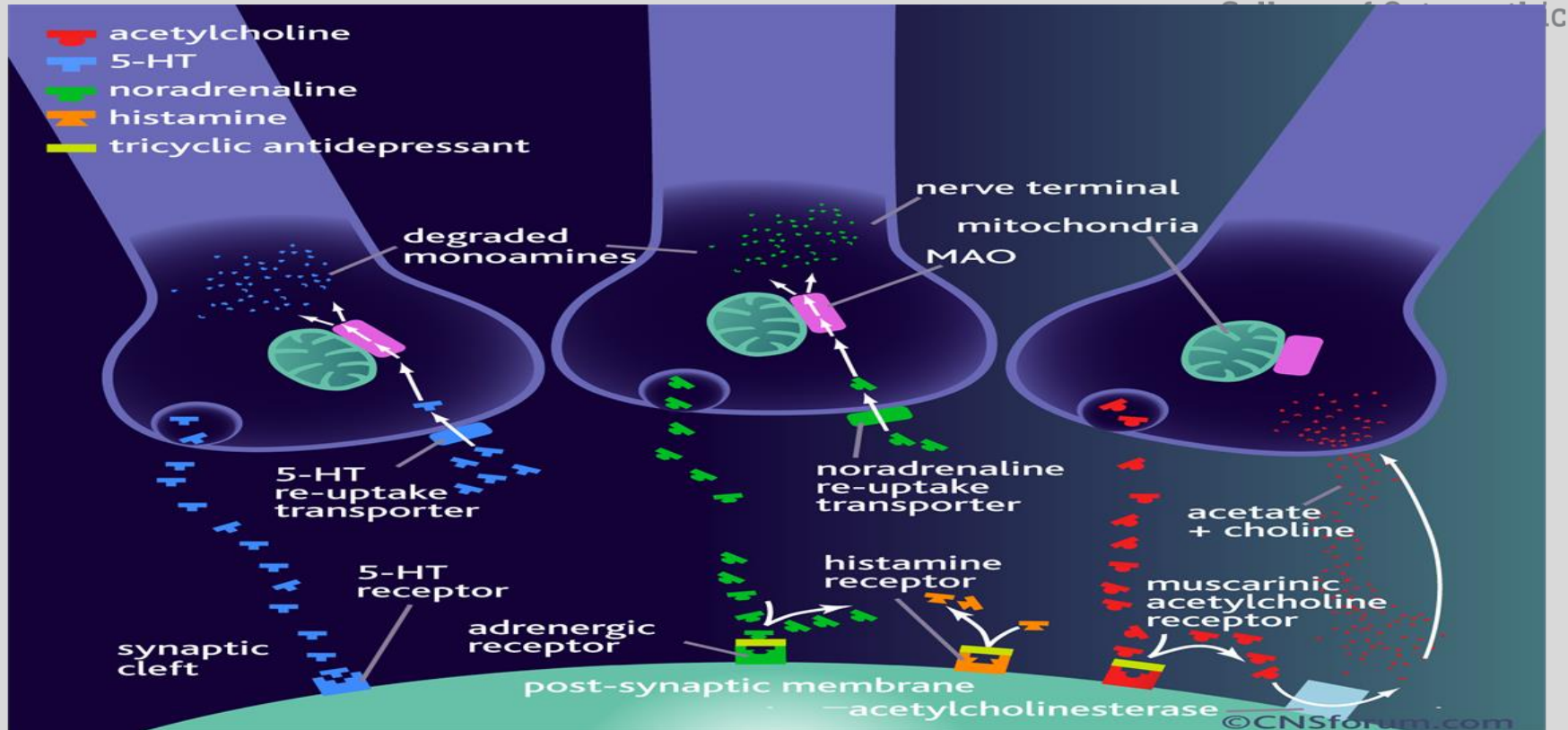
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TCA's

- Block NE and 5-HT reuptake (NET, SERT)
- Block α_1 , M, and H_1 receptors: Name those adverse effects?
- CYP drug interactions. CYP2D6 substrate. Polymorphism is associated with slow metabolism of TCAs
- Serotonin Syndrome
- 3C's: coma, convulsions, cardiotoxicity ("quinidine-like"). Administer sodium bicarbonate.

TCA's



Serotonin Syndrome

- SSRIs, TCAs, meperidine, tramadol, dextromethorphan, St. John's Wort, linezolid, "triptans"
- Symptoms: rigidity, hyperthermia, hypertension, tachycardia, myoclonus, delirium, confusion, within 24 hrs, unlike NMS. Also, GI symptoms.
- Treatment: benzodiazepines, cyproheptadine, ventilation

NMS vs Serotonin Syndrome

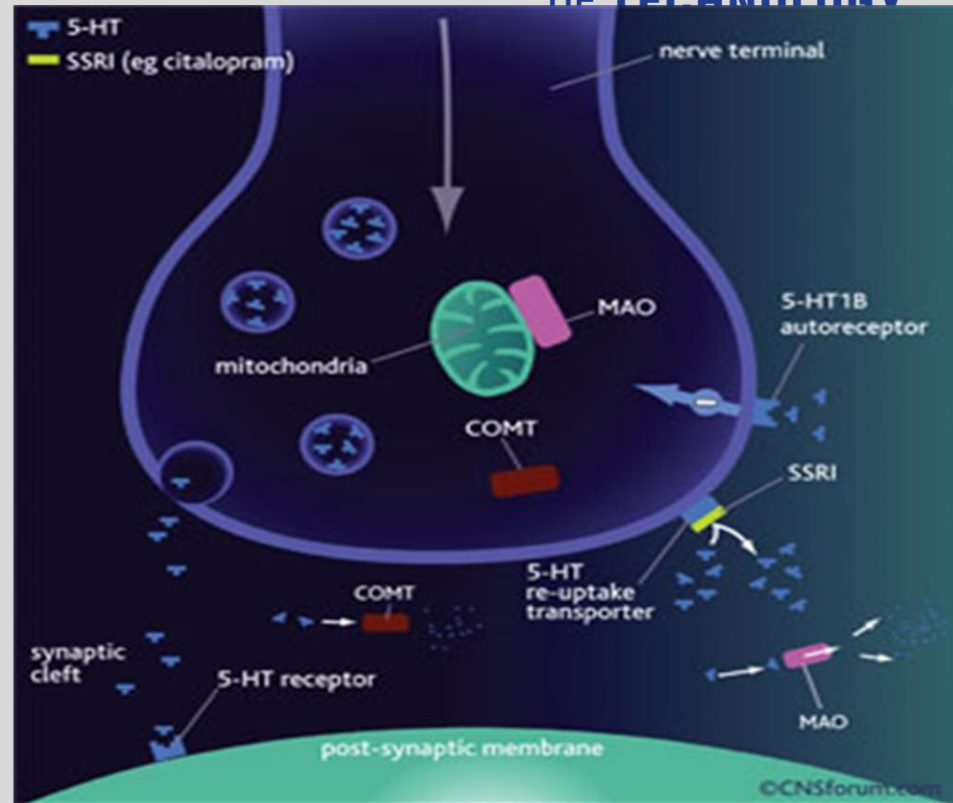
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NMS		Serotonin Syndrome
Onset	Gradual (months-weeks)	Abrupt
Findings	“Lead-pipe” rigidity	Myoclonus and tremor
Reflexes	Decreased	Increased
Pupils	Normal	Mydriasis
Drugs	Antipsychotics, halogenated anesthetics	Linezolid, St. John’s Wort, “triptans”, SSRIs, MAOIs, tricyclic antidepressants

SSRIs

- Use: depression, OCD, panic disorder, bulimia, anorexia, PTSD, premenstrual dysphoric disorder
- 5-HT specific
- Fewer ANS and sedative effects than tricyclic antidepressants
- Safer in overdose (less cardio and seizures)



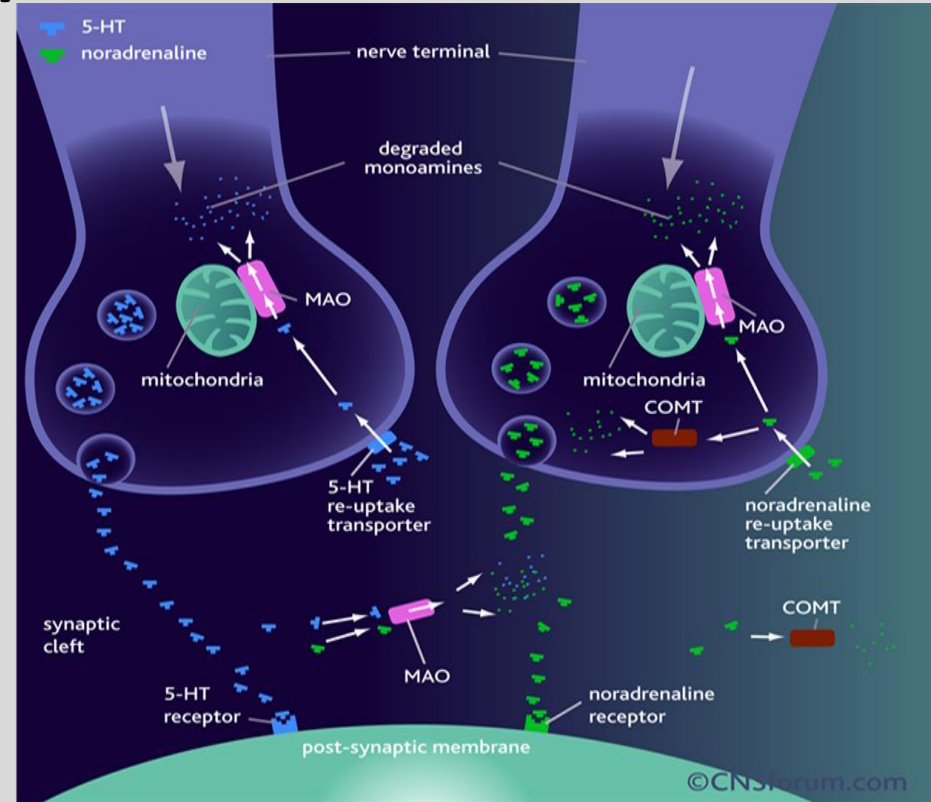
SSRI Adverse effects

- Insomnia, headache
- Nausea, vomiting (activate 5-HT₃)
- Discontinuation syndrome (least with fluoxetine)
- Bleeding abnormalities (platelets)
- Impotence (phosphodiesterase 5 inhibitor?)
- Serotonin syndrome
- Weight gain (block 5-HT_{2c})
- Drug interactions (most with fluoxetine CYP450 2D6; least with sertraline, citalopram, and escitalopram)

Serotonin and norepinephrine reuptake inhibitors (SNRI)

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- No effect on H_1 , α_1 , M receptors, unlike TCA
- Use: Depression, fibromyalgia, neuropathic pain
- Duloxetine, venlafaxine, desvenlafaxine, milnacipran

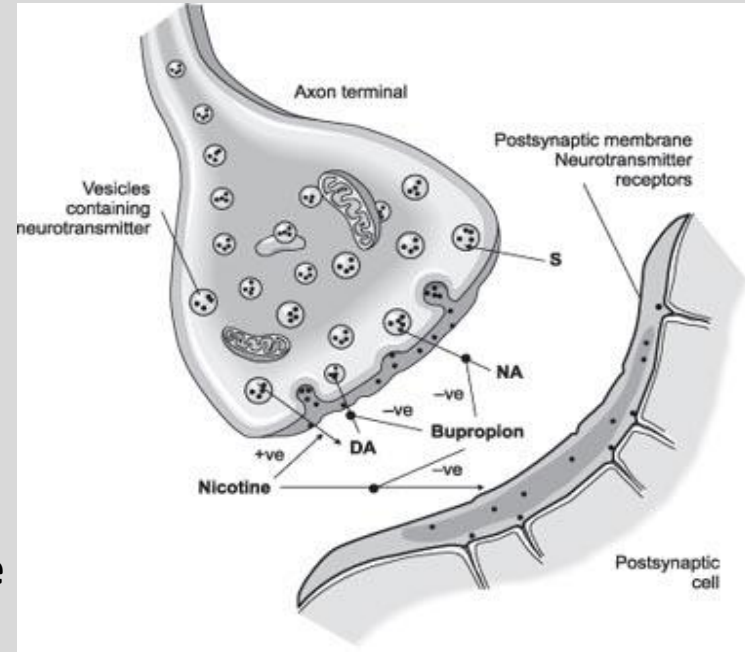


Bupropion

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- Inhibit DAT, NET, inhibits dopamine and norepinephrine reuptake.
- Little M, H₁ effects
- No sexual effects
- CYP450 2D6 inhibitor
- Dextromethorphan/bupropion (Auvelity®, 2022)
- Naltrexone/bupropion (weight loss)
- Reduce seizure threshold (not for bulimia)
- Noncompetitive antagonist at nicotinic receptor (smoking cessation)
- Varenicline: partial nicotinic receptor agonist (adverse effects: erratic behavior, suicide)

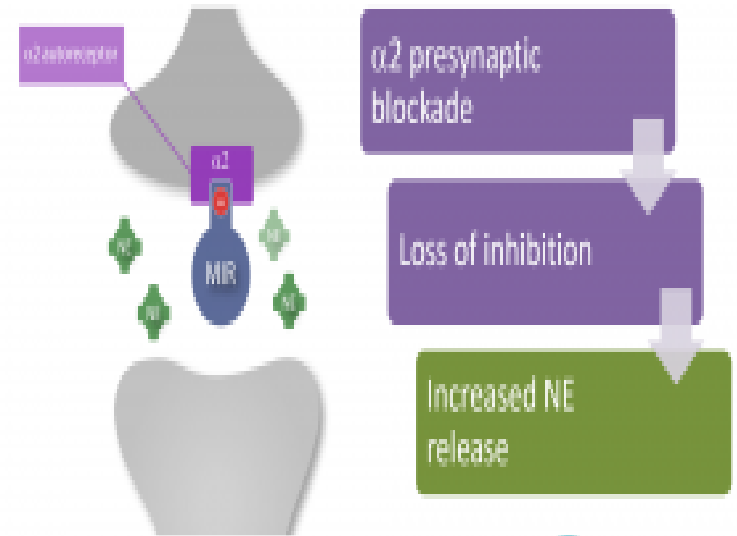


Mirtazapine

- Tetracyclic antidepressant (TeCA)
- Blocks α_2 to increase neuronal NE and 5-HT release
- Opposite **clonidine**
- Weight gain (H_1 , $5-HT_{2c}$ blockade)
Sedation (H_1 blockade)
- Blockade $5-HT_{2A}$ (no sexual dysfunction)
- Blockade $5-HT_3$ (no nausea)
- Stimulate $5-HT_{1A}$ (anxiolytic)

Source: Course Syllabus

Mirtazapine blocks α_2 presynaptic autoreceptors

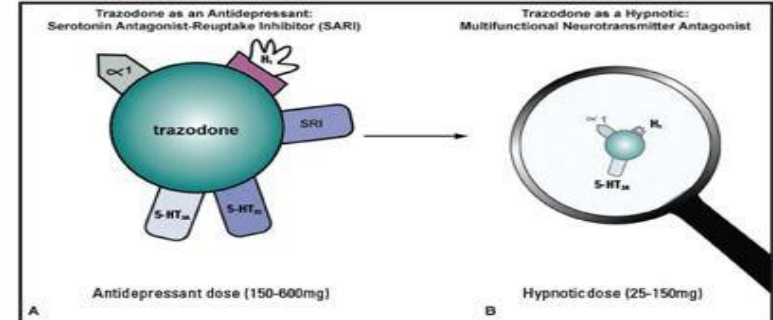


Trazodone

- Serotonin reuptake inhibitor and blocker (SARI) (5-HT_{2A})
- H₁ (sedation) and α_1 (priapism) blockade.
- Levels increased with CYP 3A4 inhibition
- Nefazodone: (hepatotoxic)
- Vilazodone (partial 5-HT_{1A} agonist).
- Vortioxetine: like vilazodone; also inhibits NET, SERT.

Source: Course Syllabus

FIGURE 3.
Trazodone as an antidepressant and as a hypnotic



A) High doses that recruit saturation of SERT at 150–600 mg of trazodone are required for antidepressant actions. At this high antidepressant dose, trazodone is a multifunctional serotonergic agent with antagonist actions at 5-HT_{2A} and 5-HT_{2C} receptors as well. Thus, its antidepressant actions are attributed to these serotonergic properties and trazodone is a multifunctional SARI. It is also an α_1 and H₁ antagonist at these doses.

B) Lower doses of trazodone in the 25–150 mg range lose saturation of SERT, and thus lose antidepressant actions while retaining multifunctional antagonist actions at 5-HT_{2A}, H₁ and α_1 receptors, and hypnotic efficacy.

α_1 = α_1 adrenergic receptor; H₁=histamine 1 receptor; SERT=serotonin reuptake inhibitor; 5-HT_{2C}=serotonin 2C receptor; 5-HT_{2A}=serotonin 2A receptor; SERT=serotonin transporter; SARI=serotonin antagonist-reuptake inhibitor.

Stahl SM. *CNS Spectr.* Vol 14, No 10. 2009.

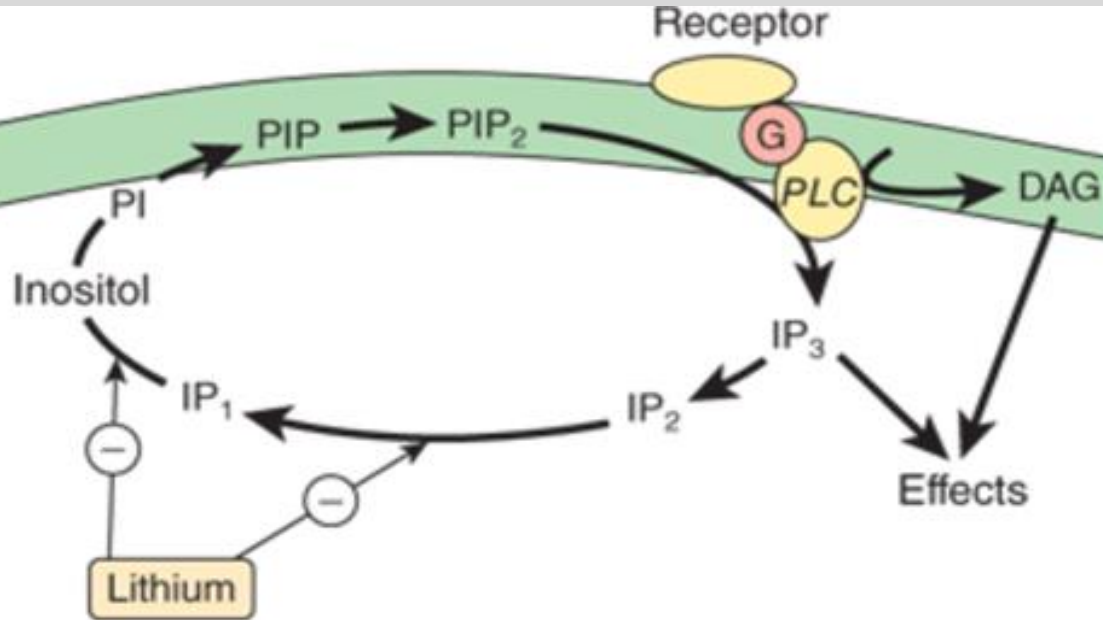
Gepirone

- Selective 5-HT_{1A} receptor agonist-- less binding to D₂ compared to buspirone.
- Less weight gain and sexual dysfunction.
- Use: Major depressive disorder, dosed once daily.
- Adverse effects: dizziness, headache, Increased QT prolongation

Bipolar disease

- Bipolar I (at least one manic episode lasting at least 7 days, then depression for at least 2 weeks)
- Bipolar II (mainly depression; with episodes of hypomania lasting several days and more mild symptoms).
- Rapid-cycling: four or more episodes in a year.
- Treatments: lithium, **valproate, carbamazepine, lamotrigine, or atypical antipsychotics**

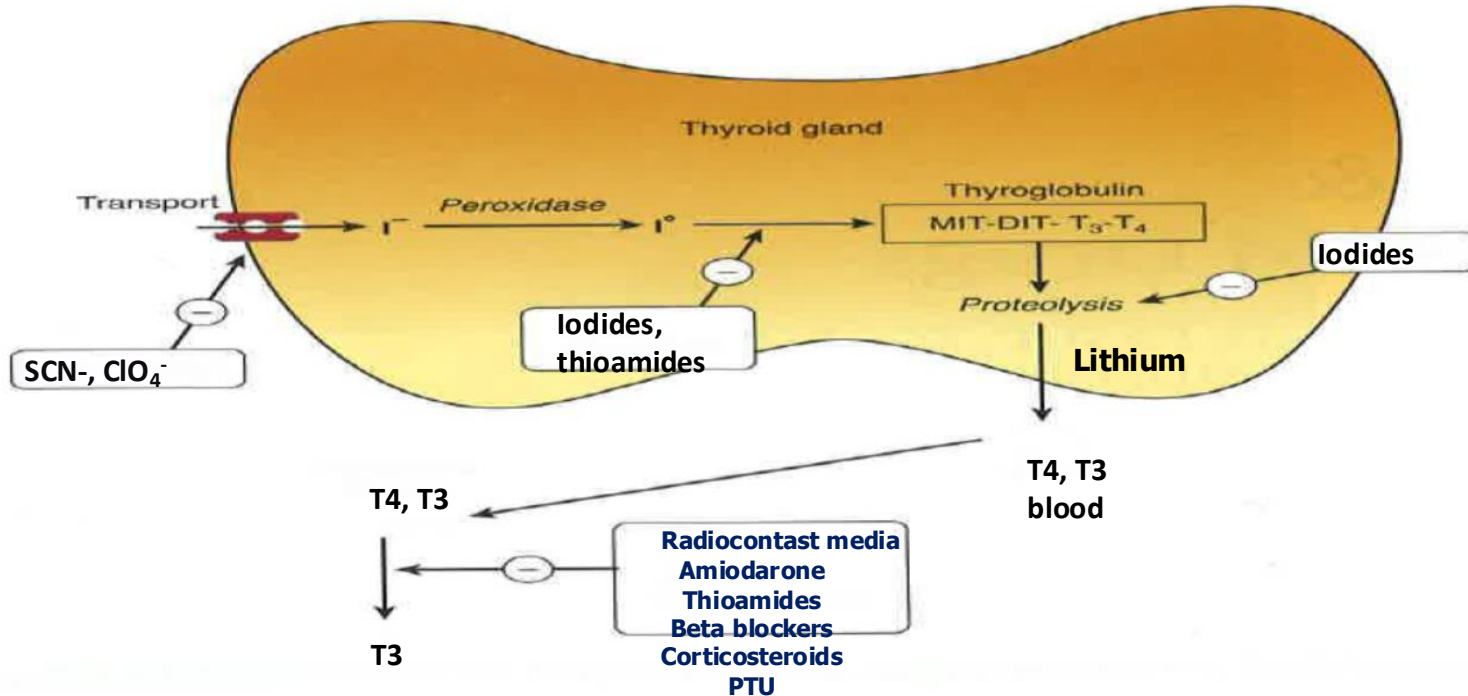
Lithium



Source: Bertram G. Katzung, Anthony J. Trevor: Basic & Clinical Pharmacology, 13th Ed.
www.accesspharmacy.com

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Iodine Organification



From: Thyroid & Antithyroid Drugs
Basic & Clinical Pharmacology, 14e, 2018

Valproic Acid

- **Broad-spectrum Mechanisms**

1. Blocks Na⁺ channels and T-type calcium channels
2. Decrease glutamate at NMDA receptors
3. Increases GABA receptor action
4. Increases GABA synthesis
5. Blocks degradation of GABA

Atypical Antipsychotics (Second generation)

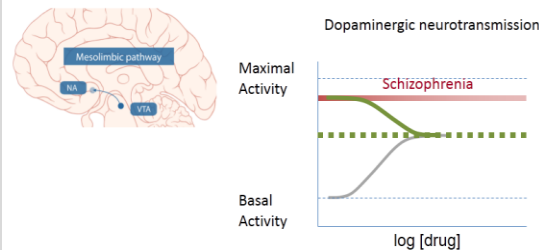
- 5-HT₂ >>>D₂; D₃ and D₄ block
- Increase DA release to relieve negative symptoms (positive too) of psychosis
- Few extrapyramidal effects (EPS) and hyperprolactinemia
- **All prolong QT interval. Change in metabolic panel.**
- Aripiprazole, olanzapine, quetiapine, risperidone, ziprasidone, and cariprazine, approved for mania

Aripiprazole

- Partial D_2 and $5-HT_{1a}$ agonist, $5-HT_{2A}$ antagonist.
- Use: bipolar disease, schizophrenia, Tourette's and irritability in autism.
- $5-HT_{2C}$ agonist (minimal weight gain).
- CYP substrate

Source: Course Syllabus

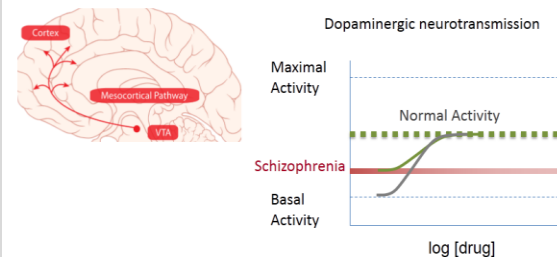
Aripiprazole therapy



Courtesy of Richard Mailman, modified from Mailman RB, Murthy V. Current pharmaceutical design 2010;16:488-501.

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Aripiprazole Therapy



Courtesy of Richard Mailman, modified from Mailman RB, Murthy V. Current pharmaceutical design 2010;16:488-501.

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Atypical Antipsychotics

- Olanzapine: Less ANS effects, Blocks D_3 , D_4 , Fewer EPS, ***weight gain***, sedation (H_1). Approved in combination with fluoxetine for mania.
 - Receptor blockade: $5-HT_{2A} > H_1 > D_4 > D_2 > \alpha_1 > D_1$
- Quetiapine: D_2 blocker, which binds for a short time. Minimal muscarinic block, but most sedation (H_1) with hypotension (α_1).
 - Receptor blockade: $H_1 > \alpha_1 > M_{1,3} > D_2 > 5-HT_{2A}$

Atypical Antipsychotics

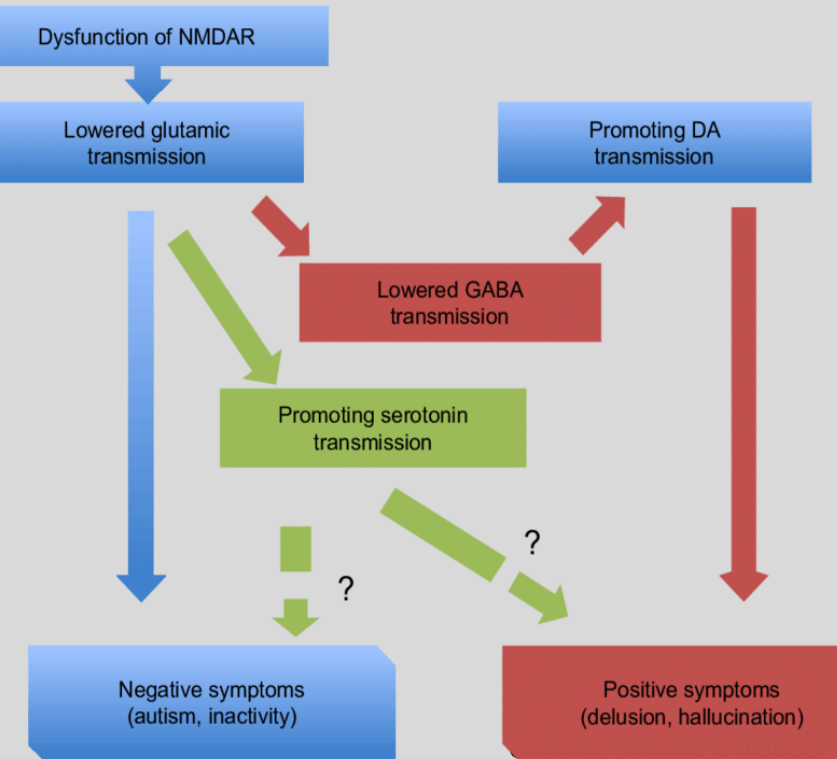
- **Risperidone: Increase prolactin, hyperlipidemia, hyperglycemia**
 - H₁ blockade (sedation, weight gain)
 - Alpha₁ block (hypotension)
- **Ziprasidone: blocks 5-HT and NE reuptake.**
 - Skin reaction, eosinophilia
 - H₁ blockade (sedation, weight gain)
 - Alpha₁ block (hypotension)
- **Cariprazine: D₂, D₃ partial agonist (5-HT_{1A} agonist)**
 - Akathisia

Lumateperone (Caplyta®)

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Glutamate hypothesis of schizophrenia

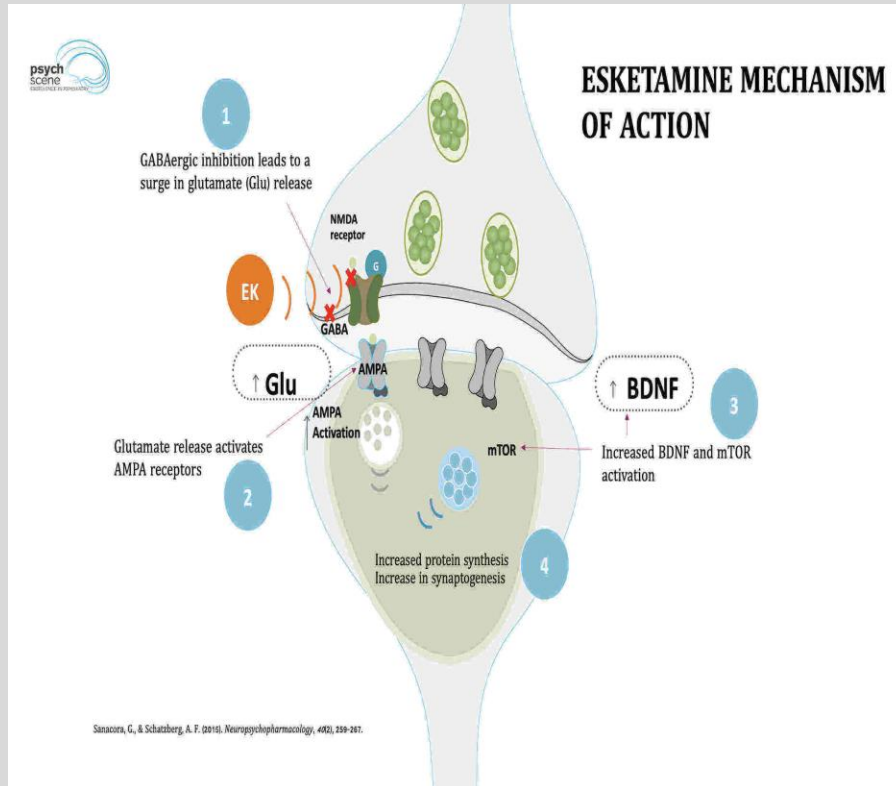


- Use: schizophrenia, MDD, bipolar I and II
- 5-HT_{2A} receptor antagonist (60x greater affinity for this receptor vs D₂ receptors).
- D₂ receptor presynaptic partial agonist and postsynaptic antagonist
“Balance”
- D₁ receptor–dependent modulator of glutamate. Augment NMDA receptors

Esketamine (Spravato®)

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- **Use:** treatment resistant depression
- **MOA:** NMDA receptor blockade
- Intranasal, administer with oral antidepressant
- Modulates dopamine. Increases BDNF
- Administered in a supervised, clinical setting. Check vitals over 2 hours.
- **AE:** Abuse liability, sedation, dissociative effects